11 The grouped frequency table gives information about the times, in minutes, that 80 office workers take to get to work.

Time (t minutes)	Frequency
$0 < t \leqslant 20$	5
$20 < t \leqslant 40$	30
$40 < t \leqslant 60$	20
$60 < t \leqslant 80$	15
$80 < t \leqslant 100$	8
$100 < t \leqslant 120$	2

(a) Complete the cumulative frequency table.

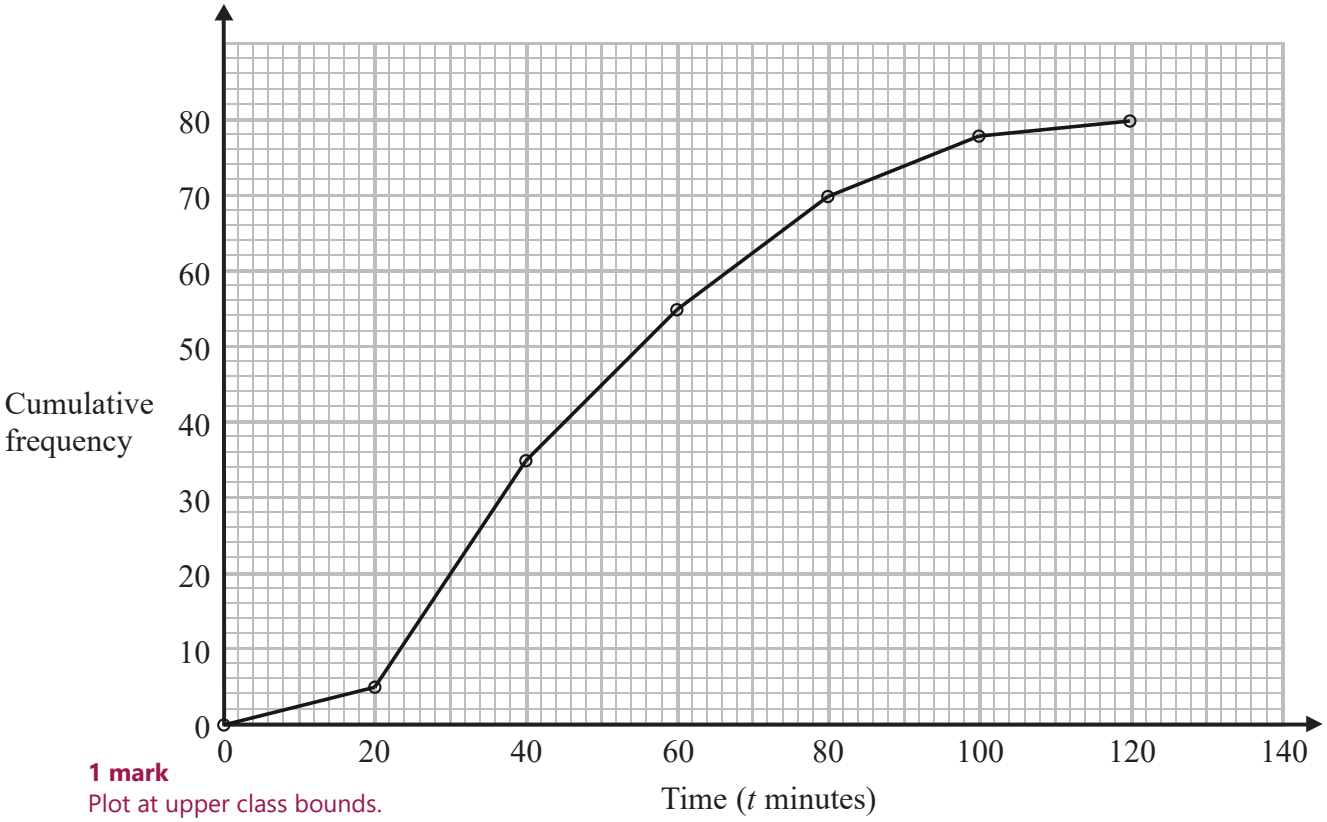
Time (t minutes)	Cumulative frequency
$0 < t \leqslant 20$	5
$0 < t \leqslant 40$	35
$0 < t \leqslant 60$	55
$0 < t \leqslant 80$	70
$0 < t \leqslant 100$	78
$0 < t \leqslant 120$	80

1 mark
Running totals:
5, 35, 55, 70, 78, 80.

(1)

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(b) On the grid, draw the cumulative frequency graph for this information.



1 mark
Plot at upper class bounds.

1 mark
Join the six points.

(2)

(c) Use your graph to find an estimate for the percentage of these office workers who take more than 90 minutes to get to work.

At 90 min, cumulative frequency ≈ 74.

Over 90 min: 80 – 74 = 6 workers.

Percentage = 6/80 × 100 = 7.5%

1 mark
Read ≈ 74 at 90 min.

1 mark
Find remainder and percentage.

1 mark
7.5%.

7.5 %

(3)

(Total for Question 11 is 6 marks)

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